

Two-Step Equations Practice

Answer Key

Grade 7 Math

Quick Answers

1. 5	4. -3	7. 19, 6	10. 5
2. 7	5. 16	8. 3, 2	11. -6, 15
3. 8	6. 7, 77	9. 15	12. 12, 45

Curriculum

Level: Grade 7 Math

7.EE.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–4 of 5 across 17 tagged checks.

1. Solve for x : $3x + 7 = 22$

Solution: Subtract 7 from both sides: $3x = 15$. Divide both sides by 3.

$\Rightarrow$

$x = 5$

2. Solve for x : $5x - 8 = 27$

Solution: Add 8 to both sides: $5x = 35$. Divide both sides by 5.

$\Rightarrow$

$x = 7$

3. Solve for x : $4x + 9 = 41$

Solution: Subtract 9 from both sides: $4x = 32$. Divide both sides by 4.

$\Rightarrow$

$x = 8$

4. Solve for x : $2x + 15 = 9$

Solution: Subtract 15 from both sides: $2x = 9 - 15 = -6$. Divide both sides by 2. A negative result is fine — the balance rules do not change.

$\Rightarrow$

$x = -3$

5. Solve for x : $\frac{x}{4} + 6 = 10$

Solution: Subtract 6 from both sides: $\frac{x}{4} = 4$. The variable is divided by 4, so undo it by multiplying both sides by 4.

$\Rightarrow$

$x = 16$

6. Solve for x , then check your solution by substituting it back into the original equation.

$$9x + 14 = 77$$

Solution: Subtract 14 from both sides: $9x = 63$. Divide both sides by 9.

$\Rightarrow$

$$x = 7$$

Check: substitute $x = 7$ into the left side: $9(7) + 14 = 63 + 14$, which equals the right side.

$\Rightarrow$

$$77$$

7. Maya solved the equation $6x - 5 = 31$ and wrote $x = 4$. Substitute $x = 4$ into the left side of her equation to find the value it really produces, then solve the equation correctly.

Solution — the check: substitute $x = 4$ into $6x - 5$: $6(4) - 5 = 24 - 5$.

$\Rightarrow$

$$19$$

Since $19 \neq 31$, Maya's value is not a solution.

Solving correctly: add 5 to both sides: $6x = 36$. Divide both sides by 6.

$\Rightarrow$

$$x = 6$$

8. Solve for x , then check your solution. $-3x + 11 = 2$

Solution: Subtract 11 from both sides: $-3x = -9$. Divide both sides by -3 ; a negative divided by a negative is positive.

$\Rightarrow$

$$x = 3$$

Check: $-3(3) + 11 = -9 + 11$, which equals the right side.

$\Rightarrow$

$$2$$

9. Solve for m : $\frac{m}{3} - 7 = -2$

Solution: Add 7 to both sides: $\frac{m}{3} = 5$. Multiply both sides by 3.

$\Rightarrow$

$$m = 15$$

10. Solve for x : $12 - 5x = -13$

Solution: Subtract 12 from both sides: $-5x = -25$. Divide both sides by -5 .

$\Rightarrow$

$$x = 5$$

11. Solve for x , then check your solution by substituting it back into the original equation.

$$-4x - 9 = 15$$

Solution: Add 9 to both sides: $-4x = 24$. Divide both sides by -4 .

$\Rightarrow$

$$x = -6$$

Check: $-4(-6) - 9 = 24 - 9$, which equals the right side.

$\Rightarrow$

$$15$$

12. Solve for x , then check your solution. $5(x - 3) = 45$

Solution: The whole bracket is multiplied by 5, so divide both sides by 5 first: $x - 3 = 9$. Then add 3 to both sides.

$\Rightarrow$

$$x = 12$$

Check: $5(12 - 3) = 5(9)$, which equals the right side.

$\Rightarrow$

$$45$$